P07 | Contribution, evidence and limits

For David Celeita and Masoud Barati | Review draft | 04 Nov 2026, 11:25 Colombia

Contribution

A sparse-PMU procedure compares new load/line interventions with the response retained after restoration. Switched DAE predictors start from estimated past states; active and originating-cause weights distinguish an intervention from its continuing effect. The electrical histories connect the intervention, its active interval and the continuing response in one sequential explanation.

Independent simulated comparison

Procedure	Timely family + source	Extra alarms
Physical bank	644/648	72
Nominal bank	643/648	92
ExtraTrees	330/648	29
ExtraTrees short	352/648	16

108 interventions replayed under three network conditions and two noise levels; 18 event parents, not 648 independent experiments. Unequal alarm operating points.

What the controlled results establish

On 18 existing continuous parents, preserving physical-state deviations gives 108/108 timely attributions and 1 extra; resetting them gives 75/108 and 166 extras. Marginal and bounded-profile scores tie on those endpoints. The state-carry benefit persists under both severity scores. A development-only continuous single-state GLR control exists; it still needs independent validation under matched information and calibration.

Difference from the hackathon and the literature

Both causal ExtraTrees models were retrained for all 18 sources on earlier simulation parents; their weights differ from the original hackathon submission. The physical gain is predominantly timely load coverage; line coverage is similar to the short-window model. Physical trajectory diagnosis and M3H histories have direct precedents. The contribution concerns recovery-consistent attribution and its operational trade-offs.

Corrections and remaining submission blockers

The manuscript now uses monitored exposure (108 s per 120-s event record); alarm-rate differences and their intervals rise by 120/108, without changing predictions or matches. It states the balanced phase-A/47-coordinate map, observation memory and measured compute cost. The active validation work tests independent continuous-observer comparison at matched information and alarm budgets, followed by cross-model transfer and calibration of attribution masses. Author approval and declarations remain open. B01 is separate and still running; no partial outcome is used here.